

Tutorial 5: Machine language

Q1. Assembling steps for instruction and \$t2, \$t1, \$t9

Step 1: identify instruction format

checking MIPS Green Card (or refer Q3 quick reference)

⇒ its R-Format instruction

⇒ R	0	rs	rt	rd	shamt	funct
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Identify operator opcode, refer to MIPS Green Card,

and $\Rightarrow 0/24_{16} = 0/36_{10}$

Step 2: mapping operands to corresponding registers number

\$t2 $\Rightarrow$ \$10, \$t1 $\Rightarrow$ \$9, \$t9 $\Rightarrow$ \$25

Refer instruction syntax,

op dst, src1, src2

$\Rightarrow$ and \$10, \$9, \$25

thus instruction is interpreted as \$10 = \$9 + \$25

Step 3: List the op/operands numbers in decimal and arrange into instruction format

Opcode = 0

rs = 9

rt = 25

rd = 10

shamt = 0 (not a shift)

funct = 36 decimal

Format	op	rs	rt	rd	shamt	funct
R	0	9	25	10	0	36

Step 4: convert into binary & corresponding hex code

Format	op						rs				rt				rd				shamt					funct															
R	0	0	0	0	0	0	0	1	0	0	1	1	1	0	0	1	0	1	0	1	0	0	0	0	0	1	0	0	1	0	0								
hex	0						1				3				9				5				0					2						4					

Hex code representation = 0x01395024

Q2. lw \$t0, 0x20(\$t7)

Step 1: identify instruction format

⇒ its I-Format instruction

I	1, 4 - 31	rs	rt	immediate (16-bit)
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Identify opcode, refer to MIPS Green Card,

lw $\Rightarrow 23_{16}$

Step 2: mapping operands to corresponding registers number.

Refer instruction syntax,

rs = \$t7 $\Rightarrow$ \$15

rt = \$t0 $\Rightarrow$ \$8

Step 3: List the op/operands numbers, arrange into instruction format and convert to hex code.

opcode				rs				rt				imm																			
23 _{hex}				15				8				20 _{hex}																			
1	0	0	0	1	1	0	1	1	1	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0
8				D				E				8				0				0				2				0			

Hex code representation = 0x8de80020

Q3. Step 1: translate machine code into binary

0x03E00008 → 0000_0011__1110_0000_0000_0000_0000_1000

Step 2: Identify the opcode and obtain corresponding instruction format

000000_11111_00000_00000_00000_001000 → 0_31_0_0_0_8 → R

Step 3: Translate (“disassemble”) to MIPS assembly instruction

Opcode/funct = 0/8₁₀ = 0/8₁₆ => check MIPS reference card to obtain corresponding instruction => jr

0x03E00008 : jr \$31 → jr \$ra

Q4.

3				C				1				1				2				5				4				3							
0	0	1	1	1	1	0	0	0	0	0	1	0	0	0	1	0	0	1	0	0	1	0	1	0	1	0	1	0	0	0	0	1	1		
F _{hex}						0 _{hex}						11 _{hex}						2543 _{hex}																	

Opcode = F_{hex} = lui

rs = 0

rt = 11_{hex} = 17₁₀ = \$s1

imm = 0x2543

Assembly code: lui \$s1, 0x2543

3				6				3				1				6				4				1				0			
0	0	1	1	0	1	1	0	0	0	1	1	0	0	0	1	0	1	1	0	0	1	0	0	0	0	0	1	0	0	0	0
D _{hex}				11 _{hex}				11 _{hex}				6410 _{hex}																			

Opcode = D_{hex} = ori

rs = 11_{hex} = 17₁₀ = \$s1

rt = 11_{hex} = 17₁₀ = \$s1

imm = 0x6410

Assembly code: ori \$s1, \$s1, 0x6410